

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12

HSC ASSESSMENT TASK 3

JUNE 2007

MATHEMATICS

EXTENSION 1

Time Allowed: 70 minutes

Name _____

Teacher _____

Instructions:

- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start each question on a new page.
- Standard integrals can be found on the last page.

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total
/8	/10	/7	/8	/10	/9	/52

Question 1

- a) Solve $2 \cos^2 x = \cos x$ for $0 \leq x \leq 2\pi$ 3
- b) Simplify $\frac{\log_m \sqrt{a}}{\log_m (a^2)}$ 2
- c) Solve $\log_e (x+1) - \log_e x = 2$. Leave your answer in exact form. 2
- d) Find $\int 3xe^{4x^2+7} dx$ 1

Question 2

- a) Find $\int \frac{6x^2}{x^3 + 4} dx$ 1
- b) Differentiate $\tan^3 x$ and hence find $\int \sec^2 x \tan^2 x dx$ 2
- c) (i) Sketch the curve $y = \log_e 2x$. Show the x intercept 1
- (ii) The area between the curve above, $y = 0$ and $y = 1$ is rotated about the y -axis. Find the generated volume in exact form. 3
- d) (i) Use a change of base to express $\log_2 5x$ in base e . 1
- (ii) Hence or otherwise, find $\frac{d}{dx}(\log_2 5x)$ 2

Question 3

- a) (i) Show that $\sin x - \cos^2 x \sin x = \sin^3 x$ 1
- (ii) Hence, and using the substitution $u = \cos x$, or otherwise, find $\int \sin^3 x dx$ 2
- b) Given the curve represented by $y = \sin^2 x$,
- (i) Sketch the curve for $-\pi \leq x \leq \pi$ 1
- (ii) Find the total area between the x -axis and the curve above 3

Question 4

- a) The function f is defined as $y = x(x - 2)$.
- (i) Sketch f and state the largest positive domain for which an inverse f^{-1} exists. 2
- (ii) Sketch f^{-1} . Show two key points 1
- (iii) Find the coordinates of the point where f and f^{-1} intersect 1
- b) Explain, without evaluating, why $\sin^{-1}(\sin \frac{3\pi}{4}) \neq \frac{3\pi}{4}$ 1
- c) (i) Write the expansion of $\tan(\theta - \alpha)$ 1
- (ii) Hence or otherwise, express $\tan[\cos^{-1}(-x)]$ in terms of x only 2

Question 5

- a) Differentiate $y = \tan^{-1}(\sin 2x)$ 2
- b) Consider the function $f(x) = \cos^{-1}(x^2)$
- (i) Write the domain and range of $y = f(x)$ 2
- (ii) Find the slope of the tangent where the curve crosses the y axis. 2
- (iii) Sketch the curve $y = f(x)$ 1
- c) Use the expansion of $\sin(A + B)$ to express $\sin^{-1}(\frac{4}{5}) + \sin^{-1}(\frac{12}{13})$ in the form $\sin^{-1} M$. 3

Question 6

- a) Find $\int \frac{dx}{\sqrt{9 - 4x^2}}$ 2
- b) (i) Find $\frac{d}{dx}(x \tan^{-1} x)$ 1
- (ii) Hence, and using a suitable rearrangement, evaluate $\int_0^1 \tan^{-1} x \, dx$ 3
- c) Using a diagram, or otherwise, evaluate $\int_0^1 \sin^{-1} x \, dx$. Give your answer in exact form. 3

SOLUTIONS

① a) $2\cos^2 x - \cos x = 0$

$$\cos x(2\cos x - 1) = 0$$

$$\cos x = 0 \text{ or } \frac{1}{2} \quad \leftarrow \textcircled{1}$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{3}, \frac{5\pi}{3} \quad \leftarrow \textcircled{1} \quad \leftarrow \textcircled{1}$$

b) $\frac{\frac{1}{2} \log_m a}{2 \log_m a} = \frac{1}{4} \quad \leftarrow \textcircled{1}$

c) $\log_e \left(\frac{x+1}{x} \right) = 2 \quad \leftarrow \textcircled{1}$

$$\frac{x+1}{x} = e^2$$

$$x+1 = xe^2$$

$$x(1-e^2) = -1$$

$$x = \frac{-1}{1-e^2} \text{ or } \frac{1}{e^2-1} \quad \textcircled{1}$$

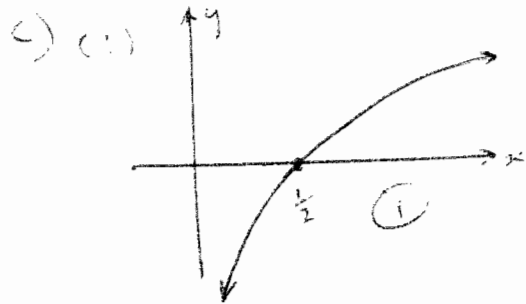
d) $\frac{3}{8} \int 8x e^{4x^2+7} dx$
 $= \frac{3}{8} e^{4x^2+7} + c \quad \textcircled{1}$

② a) $2 \int \frac{3x^2}{x^3+4} dx$
 $= 2 \log(x^3+4) + c \quad \textcircled{1}$

b) $\frac{d}{dx}(\tan^3 x) = 3 \tan^2 x \sec^2 x \quad \textcircled{1}$

$$\int \sec^2 x \tan^2 x dx$$

$$= \frac{1}{3} \tan^3 x + c \quad \textcircled{1}$$



(ii) $y = \log_e 2x \Rightarrow 2x = e^y$
 $x = \frac{1}{2} e^y$

$$\therefore \text{Vol} = \pi \int_0^1 \left(\frac{1}{2} e^y \right)^4 dy \quad \textcircled{1}$$

$$= \frac{\pi}{4} \int_0^1 e^{2y} dy$$

$$= \frac{\pi}{4} \left[\frac{1}{2} e^{2y} \right]_0^1 \quad \textcircled{1}$$

$$= \frac{\pi}{8} (e^2 - e^0)$$

$$= \frac{\pi}{8} (e^2 - 1) u^3 \quad \textcircled{1}$$

d) (i) $\log_2 5x = \frac{\log_e 5x}{\log_e 2}$
 $\textcircled{1} \log_e 2$

(ii) deriv. = $\frac{\frac{5}{5x}}{\log_e 2} \quad \textcircled{1}$

$$= \frac{1}{x \log_e 2} \quad \textcircled{1}$$

3

a) i) $\sin x (1 - \cos^2 x) = \sin x \sin^2 x$
 $= \sin^3 x$

ii) $\int \sin^3 x dx = \int (\sin x - \cos^2 x \sin x) dx$

$u = \cos x$
 $\frac{du}{dx} = -\sin x$
 $dx = \frac{du}{-\sin x}$

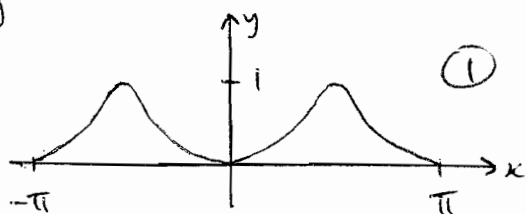
$= \int \sin x (1 - \cos^2 x) dx$
 $= \int \cancel{\sin x} (1 - u^2) \frac{du}{-\cancel{\sin x}}$

$= \int (u^2 - 1) du$

$= \frac{u^3}{3} - u + C$

$= \frac{\cos^3 x}{3} - \cos x + C$

b) (i)



(ii)

$A = 2 \int_0^\pi \sin^2 x dx$

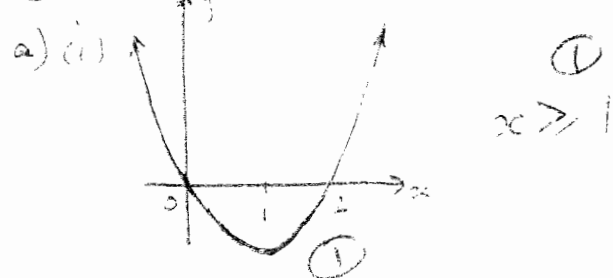
$= 2 \int_0^\pi \frac{1}{2} (1 - \cos 2x) dx$

$= \left[x - \frac{\sin 2x}{2} \right]_0^\pi$

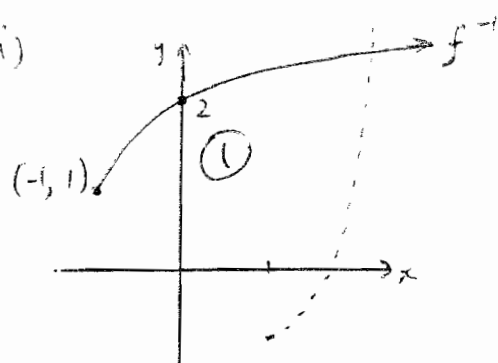
$= (\pi - 0) - (0 - 0)$

$= \pi$

4



(ii)



(iii) intersect on $y = x$

$\therefore x(x-2) = x$

$\therefore x^2 - 2x - x = 0$

$\therefore x(x-3) = 0$

$\therefore x = 0 \text{ or } 3$

$\therefore$ intersect at $(3, 3)$

b) Range of $\sin^{-1} m$ is

$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

c) (i) $\tan(\theta - \alpha) = \frac{\tan \theta - \tan \alpha}{1 + \tan \theta \tan \alpha}$

(ii)

$\tan[\cos^{-1}(x)] = \tan(\pi - \cos^{-1} x)$

$= \tan(\pi - \alpha)$

$= \tan \pi - \tan \alpha$

$= 0 - \frac{\sqrt{1-x^2}}{x}$

$= \frac{0 - \sqrt{1-x^2}}{1 + 0}$

$= -\sqrt{1-x^2}$

5

$$\begin{aligned} \text{a) } \frac{dy}{dx} &= \frac{1}{1+(\sin 2x)^2} \times \cos 2x \times 2 \\ &= \frac{2 \cos 2x}{1 + \sin^2 2x} \quad \text{--- ①} \end{aligned}$$

$$\text{b) (i) } -1 \leq x^2 \leq 1$$

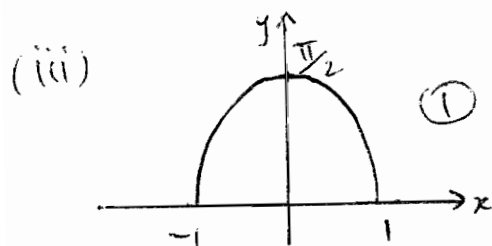
$$\therefore 0 \leq x^2 \leq 1$$

$$\therefore D: -1 \leq x \leq 1 \quad \text{①}$$

$$R: 0 \leq y \leq \frac{\pi}{2} \quad \text{①}$$

$$\begin{aligned} \text{(ii) } \frac{dy}{dx} &= \frac{-1}{\sqrt{1-(x^2)^2}} \times 2x \\ &= \frac{-2x}{\sqrt{1-x^4}} \quad \text{①} \end{aligned}$$

When $x=0$, slope of tangent = 0



$$\text{c) Let } A = \sin^{-1} \frac{4}{5}, B = \sin^{-1} \frac{12}{13}$$

$$\therefore \sin A = \frac{4}{5}, \sin B = \frac{12}{13}$$

$$\sin(A+B) = \sin A \cos B + \sin B \cos A$$

$$\begin{aligned} \begin{array}{c} \text{5} \\ \text{A} \quad \text{4} \\ \text{3} \end{array} &= \frac{4}{5} \times \frac{5}{13} + \frac{12}{13} \times \frac{3}{5} \quad \text{①} \\ &= \frac{20}{65} + \frac{36}{65} \\ &= \frac{56}{65} \end{aligned}$$

$$\therefore A+B = \sin^{-1} \left(\frac{56}{65} \right) \quad \text{①}$$

6

$$\begin{aligned} \text{a) } \int \frac{dx}{\sqrt{9-4x^2}} &= \int \frac{dx}{\sqrt{4} \sqrt{\frac{9}{4}-x^2}} \\ &= \frac{1}{2} \int \frac{dx}{\sqrt{\left(\frac{3}{2}\right)^2-x^2}} \end{aligned}$$

① method

① answer

$$= \frac{1}{2} \sin^{-1} \left(\frac{x}{\frac{3}{2}} \right)$$

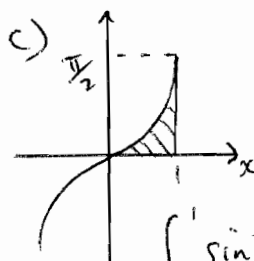
$$= \frac{1}{2} \sin^{-1} \frac{2x}{3} + C$$

b) (i)

$$\begin{aligned} \frac{d}{dx} (x \tan^{-1} x) &= 1 \times \tan^{-1} x + \frac{1}{1+x^2} \times x \\ &= \tan^{-1} x + \frac{x}{1+x^2} \quad \text{①} \end{aligned}$$

(ii)

$$\begin{aligned} \int_0^1 \tan^{-1} x &= \int_0^1 \frac{d}{dx} (x \tan^{-1} x) - \int_0^1 \frac{x}{1+x^2} \quad \text{①} \\ &= \left[x \tan^{-1} x \right]_0^1 - \left[\frac{1}{2} \log(1+x^2) \right]_0^1 \quad \text{①} \\ &= \tan^{-1} 1 - 0 - \frac{1}{2} (\log 2 - \log 1) \\ &= \frac{\pi}{4} - \frac{1}{2} \log 2 \quad \text{①} \\ &\quad (\text{approx. } 0.4) \end{aligned}$$



$$\begin{aligned} \int_0^1 \sin^{-1} x \, dx &= \left(\frac{\pi}{2} \times 1 \right) - \int_0^{\frac{\pi}{2}} (\sin y) \, dy \quad \text{①} \\ &= \frac{\pi}{2} - \left[-\cos y \right]_0^{\frac{\pi}{2}} \quad \text{①} \\ &= \frac{\pi}{2} + (\cos \frac{\pi}{2} - \cos 0) \\ &= \frac{\pi}{2} + 0 - 1 \end{aligned}$$